



FLEET AND SHORE STAFF TRAINING

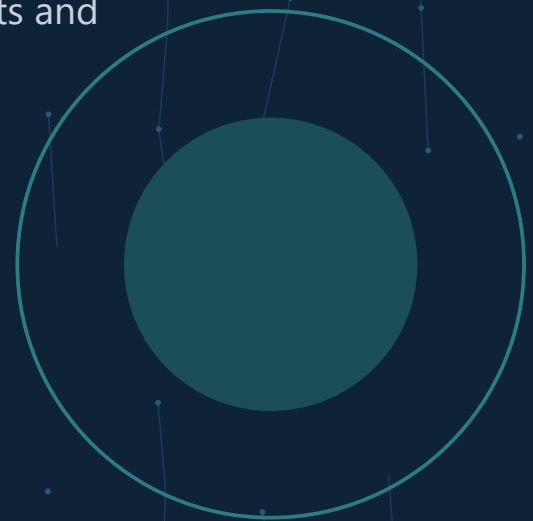
COLREG Training

International Regulations for Preventing Collisions at Sea (1972, as amended) — steering and sailing rules, lights and shapes, sound and light signals.

IMO COLREG 1972

STCW Table A-II/1

Company SMS



What We Will Cover

01

General Provisions

Application, responsibility and core definitions

02

Rules for Any Condition of Visibility

Look-out, safe speed, risk of collision, action to avoid collision

03

Narrow Channels and Separation Schemes

Rule 9 and Rule 10 — traffic separation schemes

04

Vessels in Sight of One Another

Head-on, crossing, overtaking — give-way / stand-on vessel

05

Conduct in Restricted Visibility

Rule 19 — acting on radar contact, six restrictions

06

Lights and Shapes

Rule 20–31 — which vessel carries which light

07

Sound and Light Signals

Rule 32–37 — equipment, manoeuvring, fog and distress signals

08

Exemptions and Summary

Rule 38 and a quick recap of the whole course



PART 1

General Provisions

Where the rules apply, the limits of responsibility, and the core definitions you will meet again and again.



Application and Responsibility

COLREG applies to all vessels on the high seas; but complying with the rules is never an excuse to ignore good seamanship or the special circumstances of a case.



Rule 1 – Application

Applies to all vessels upon the high seas and all waters connected therewith, so far as local rules (harbour/river) do not conflict.



Rule 2(a) – Responsibility

Nothing exonerates any vessel, owner, master or crew from the consequences of neglect.



Rule 2(b) – Special Circumstances

When departing from the rules is necessary to avoid immediate danger, due regard is given to good seamanship and the special circumstances of the case.



The Golden Rule

"I followed the rules" does not mean "I am not responsible for the collision" — common sense always outranks the letter of the rule.

RULE 3 – GENERAL DEFINITIONS

Terms You Must Know

These definitions are the vocabulary every later rule is built on.



Vessel

Every description of water craft used or capable of being used as a means of transportation on water — including seaplanes.



Power-Driven Vessel / Sailing Vessel

Any vessel propelled by machinery is a “power-driven vessel”; a “sailing vessel” is one under sail alone, even if fitted with machinery, provided it is not being used.



Vessel Engaged in Fishing

A vessel fishing with nets, lines, trawls or other fishing apparatus that restricts manoeuvrability — but not a vessel fishing with trolling lines.



Vessel Not Under Command (NUC)

A vessel which, through some exceptional circumstance, is unable to manoeuvre as required and therefore cannot keep out of the way of another vessel.



PART 2

Rules for Any Condition of Visibility

Rules that apply at all times, in clear weather or fog: look-out, speed, risk assessment and action to avoid collision.



Look-out and Safe Speed

These two rules are the precondition for every collision-avoidance decision that follows.



Rule 5 – Look-out

Every vessel shall at all times maintain a proper look-out by sight and hearing, and by all available means appropriate to the circumstances (radar included), so that a full appraisal of the situation and risk of collision can be made.



Rule 6 – Safe Speed

Speed must allow effective action to avoid collision and stopping within a distance appropriate to the circumstances. Visibility, traffic density, manoeuvrability, background lights at night, wind/sea/current, and the state of water depth relative to draught are all considered.



Extra Factors When Using Radar

Radar equipment limitations, range-scale setting, the possibility that small targets (boats, ice) may not be detected, and the number of targets that can complicate assessment.



Practical Consequence

Safe speed is not a fixed number — 18 knots may be safe in open water, while even 6 knots may be too fast in thick fog in a narrow strait.

RULE 7 – RISK OF COLLISION

How to Recognise the Risk

If in doubt, assume risk exists.



Compass Bearing Not Changing

If the compass bearing of an approaching vessel does not appreciably change, risk of collision shall be deemed to exist — especially when approaching a very large vessel or a tow.



Early and Systematic Radar Use

Obtain early warning through long-range scanning; looking is not enough — systematic plotting (ARPA/manual) is essential.



Never Assume From Scanty Information

Assumptions shall never be made on the basis of scanty information; when in doubt, risk is deemed to exist.



Change at Close Range

Even if the bearing does change, risk may still exist when approaching a very large vessel or a tow, or at very close range.

What a Good Manoeuvre Looks Like

This rule gives the golden rules for HOW to manoeuvre.



Early and Substantial

Any action taken shall be positive, made in ample time and with due regard to good seamanship — a succession of small changes creates confusion.



Obvious on Radar Too

A change of course/speed must be large enough to be readily apparent to another vessel observing visually or by radar.



Slowing Down / Stopping

A course alteration alone may be the most effective action if there is sea room; otherwise slacken speed, stop, or reverse propulsion.



Leave Enough Room

Action taken must not create a new close-quarters situation for other vessels; the situation must be checked until the other vessel is finally past and clear.

RULE 9 – NARROW CHANNELS

Conduct in Channels and Fairways

The obligation to keep to the starboard edge, and overtaking priorities.



Keep to Starboard

In a narrow channel or fairway, keep as near to the outer limit on your starboard side as is safe and practicable.



Small / Fishing Vessels

Vessels under 20m, sailing vessels and fishing vessels must not impede a vessel that can safely navigate only within the channel.



Do Not Cross the Channel

No vessel shall cross a narrow channel if this impedes a vessel that can only navigate safely within that channel.



Sound Agreement (Overtaking in a Channel)

Overtaking in a narrow channel requires the permission of the vessel being overtaken: two prolonged plus one short blast = intend to overtake on her starboard, two prolonged plus two short = on her port; agreement is one prolonged-one short-one prolonged-one short.

RULE 10 – TRAFFIC SEPARATION SCHEME (TSS)

Conduct Between the Lanes

Special rules that apply within IMO-adopted separation schemes.



Proceed With the General Flow

In a traffic lane, proceed in the general direction of traffic flow for that lane; keep clear of the separation line or zone as far as practicable.



Crossing a Lane

If you must cross a traffic lane, do so on a heading as nearly as practicable at right angles to the general flow direction.



Inshore Traffic Zone

Vessels under 20m, sailing vessels and fishing vessels may use the inshore traffic zone; larger vessels should avoid it unless necessary.



Things to Avoid

Enter or leave a lane at its terminations where practicable; do not impede traffic flow through fishing near the scheme.



PART 3

Vessels in Sight of One Another

Between two vessels that can see each other, who gives way and who holds course — the most-used and most-confused part of COLREG.



Application and Sailing Vessels

This section only applies to vessels that can see one another (radar contact alone is not enough).



Rule 11 – Application

The rules in this section (12–18) apply only to vessels in sight of one another.



Rule 12(a)(i)

When each has the wind on a different side, the vessel with the wind on the port side keeps out of the way of the other.



Rule 12(a)(ii)-(iii)

When both have the wind on the same side, the vessel to windward keeps out of the way of the vessel to leeward; if the side the wind is on cannot be determined, a vessel is deemed to have the wind on the same side as herself.



Rule 12(b)

The windward side is the side opposite to that on which the mainsail is carried.

RULE 13

Overtaking

The overtaking rule applies to any type of vessel.

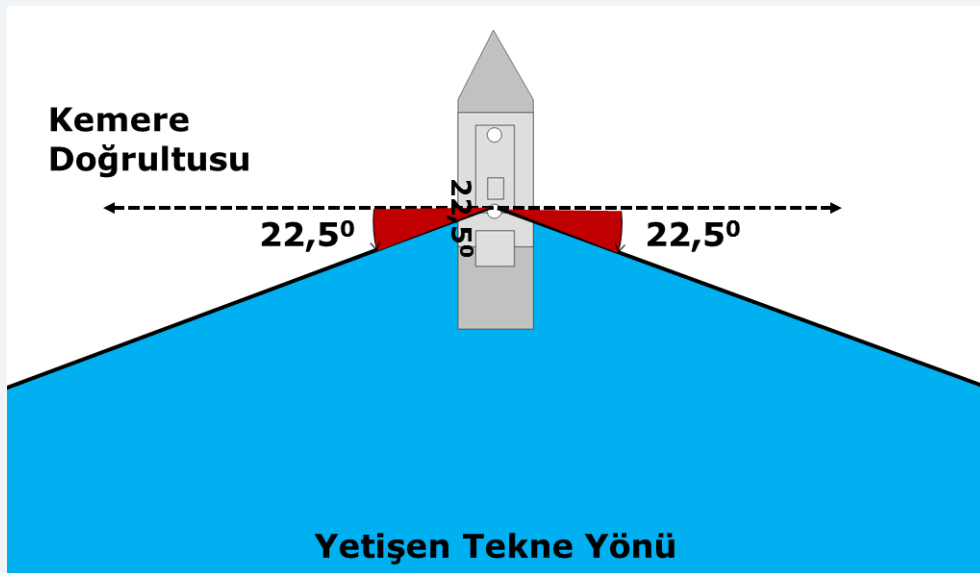


Figure 2.2 — Overtaking Vessel Bearing (Source: Turkish MoNE maritime module (diagram labels are in the original Turkish))

Definition

A vessel is OVERTAKING when approaching another from more than 22.5° abaft her beam. At night, if only the sternlight is visible (no sidelights), an overtaking situation is assumed.

Rule

The overtaking vessel keeps clear of the vessel being overtaken until finally past and clear. This overrides the whole in-sight-of-one-another hierarchy.

Once Overtaking, Always Overtaking

In case of doubt, a vessel shall assume she is overtaking; even if the bearing later changes, the situation does not change until the overtaking manoeuvre is complete.

RULE 14

Head-on Situation

Two power-driven vessels meeting on reciprocal or nearly reciprocal courses.

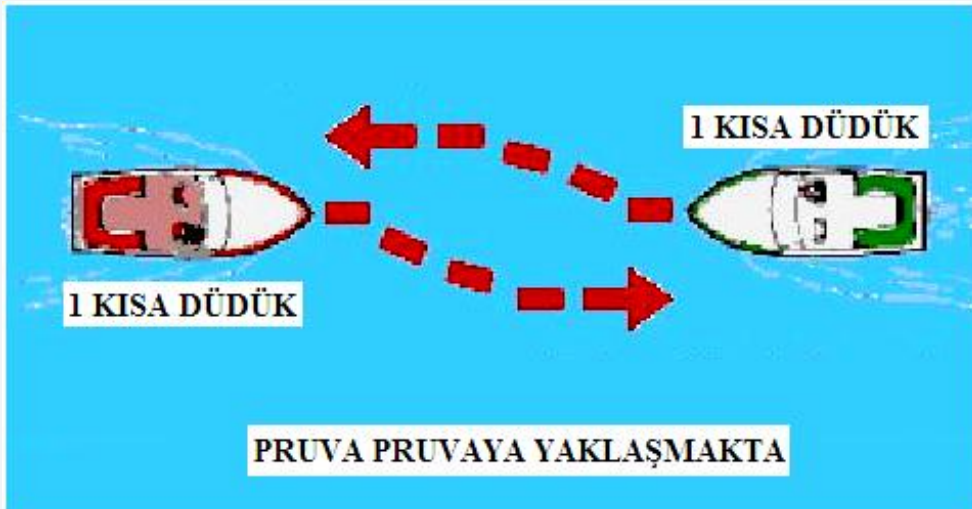


Figure 2.3 — Approaching Head-on (Source: Turkish MoNE maritime module (diagram labels are in the original Turkish))

Recognition

It is a head-on situation when each vessel sees the other's masthead lights in a line or nearly in a line, and/or sees both sidelights of the other.

Rule

EACH vessel alters course to STARBOARD, so that each passes on the PORT side of the other (port-to-port). If the course change is made at close range, one short blast is sounded.

When in Doubt

If there is any doubt whether such a situation exists, it shall be assumed to exist and acted upon accordingly.

RULE 15

Crossing Situation

The courses of two power-driven vessels cross such that a risk of collision may exist.

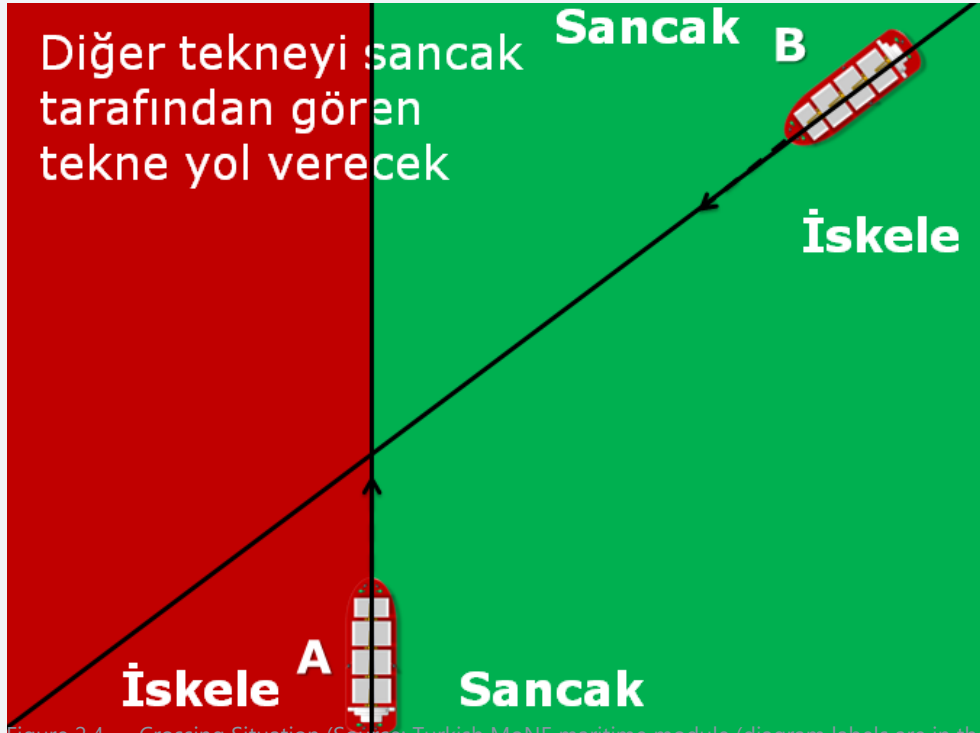


Figure 2.4 — Crossing Situation (Source: Turkish MoNE maritime module (diagram labels are in the original Turkish))

Who Gives Way

The vessel which has the other on her OWN STARBOARD side (green zone) shall keep out of the way and, if circumstances permit, avoid crossing ahead of the other.

Who Holds Course

The vessel which has the other on her own PORT side (red zone) is the stand-on vessel; under Rule 17 she keeps her course and speed.

Rule of Thumb

As the source book puts it: "See red, alter course; see green, carry on." If you appear on the other vessel's starboard side, YOU are the give-way vessel.

Give-way Vessel and Stand-on Vessel

Both sides carry firm duties — the stand-on vessel is not a passive party.



Rule 16 – Give-way Vessel

Shall take early and substantial action to keep well clear — a small last-minute correction is not enough.



Rule 17(a)(i) – Stand-on Vessel

Keeps her course and speed — she does not manoeuvre suddenly and create confusion.



Rule 17(a)(ii) – Action Alone

If the give-way vessel is NOT taking appropriate action, the stand-on vessel may take action alone to avoid collision.



Rule 17(b)-(c) – Last Resort

When collision cannot be avoided by the give-way vessel's action alone, the stand-on vessel also takes the best aid to avoiding collision; in a crossing situation she shall not alter course to PORT for a vessel on her own port side.

Responsibilities Between Vessels

The vessel with greater manoeuvrability is responsible for keeping clear.

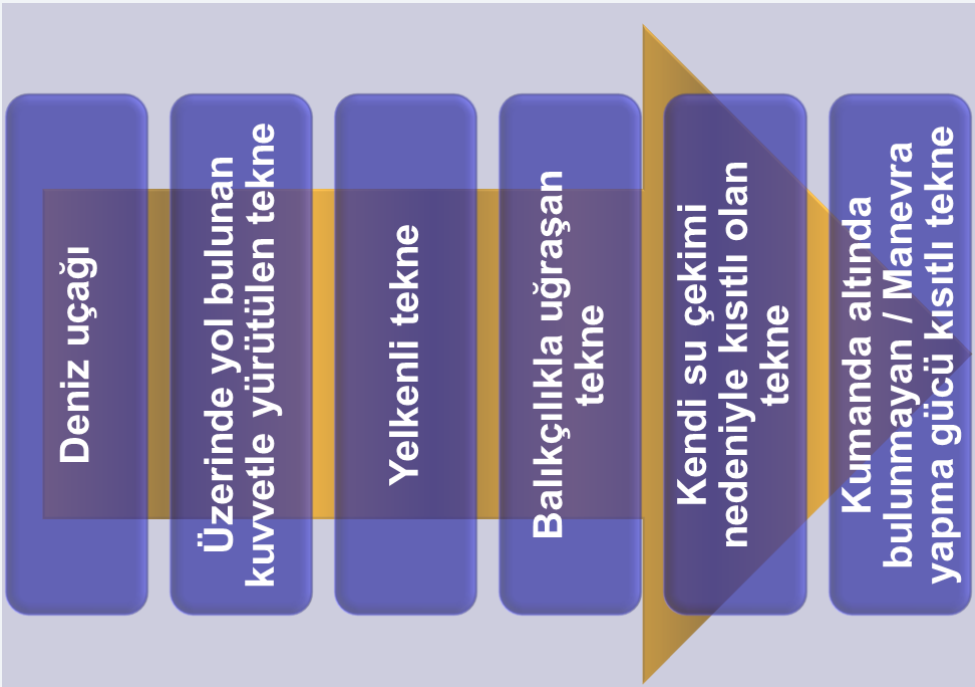


Figure 2.8 — Responsibilities Between Vessels (Source: Turkish MoNE maritime module (diagram labels are in the original Turkish))

Top of the List: Not Under Command / Restricted in Ability to Manoeuvre

These two vessel types outrank everything else — every other vessel keeps out of their way.

Next: Constrained by Her Draught

A vessel constrained by her draught is given priority over a vessel engaged in fishing and over a sailing vessel.

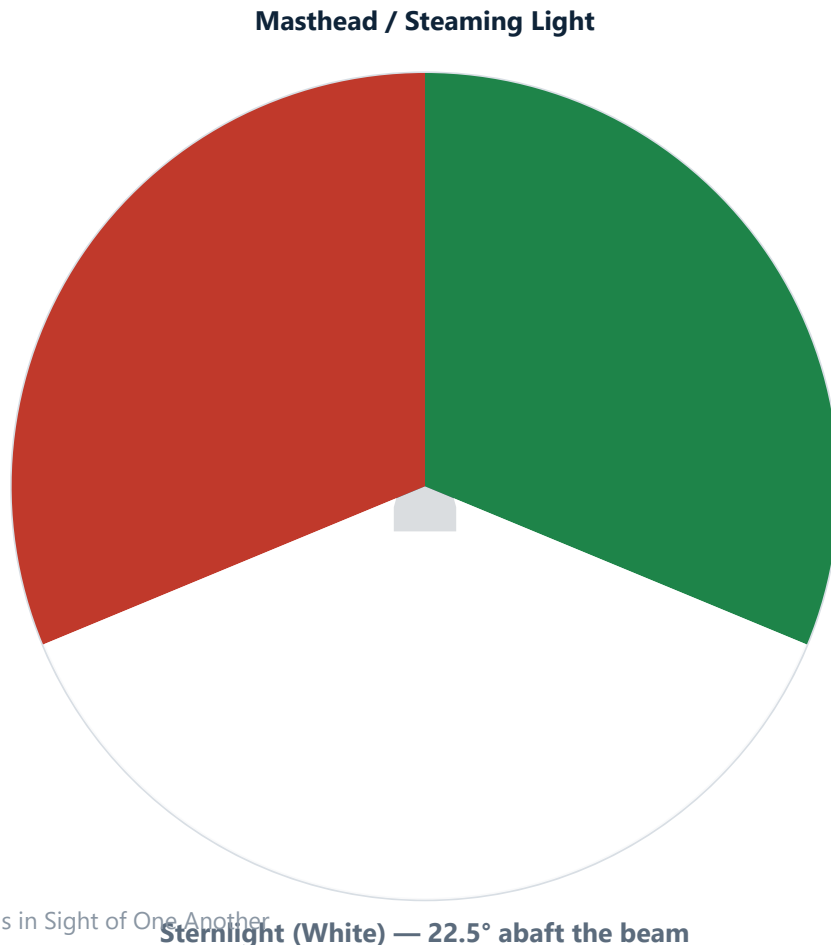
Bottom of the List: Power-Driven Vessel and Seaplane

At the bottom of the source book's order — the vessels that give way to everyone else — sit the ordinary power-driven vessel and the seaplane.

RULE 21 – DEFINITIONS

Light Visibility Sectors

Each light is visible only within a specific arc relative to the vessel's heading; these arcs are the key to recognising which vessel you are seeing, and from which side.



Why it matters

- A vessel that sees a green (starboard) sidelight is being seen from that vessel's starboard side.
- If you see only the white sternlight, you are overtaking that vessel (Rule 13).
- If you see both red and green sidelights together, the vessel is heading straight at you (head-on situation, Rule 14).
- These sectors are the foundation of the “recognise the situation from the light you see” logic in Rules 13/14/15.



PART 4

Conduct in Restricted Visibility

The special rule that applies when fog, rain, snow or darkness removes mutual visibility.

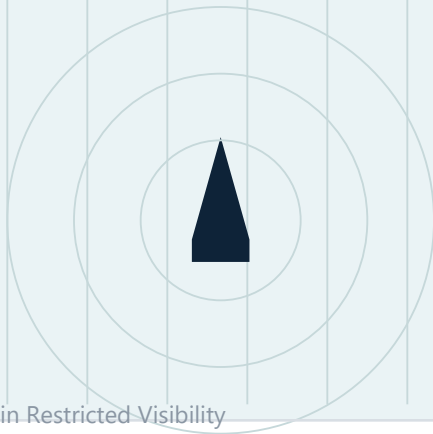


RULE 19

Conduct in Restricted Visibility

You have a radar contact but no visual sighting — the rules of Part 3 no longer apply.

Fog / Restricted Visibility — target visible on radar only



Safe Speed + Engines Ready

Proceed at a safe speed adapted to the circumstances; keep engines ready for immediate manoeuvre.

FORBIDDEN Actions on Radar Detection

Do not alter course to PORT for a vessel forward of the beam (unless overtaking); do not alter course toward a vessel abeam or abaft the beam.

On Hearing a Fog Signal

If you hear a fog signal forward of the beam, or cannot avoid a close-quarters situation with a target forward of the beam, reduce speed to the minimum at which you can keep on your course, and if necessary take all way off.

Common Mistakes

Rule 19 causes the most dangerous errors when confused with Part 3 reflexes.



Reaching for the Rule 15 Reflex

"She sees me on her starboard side, so I give way" does NOT apply in restricted visibility — that logic only applies with visual contact.



Early Plotting Is Essential

CPA/TCPA must be calculated early with ARPA/manual plotting; a target that seems "far enough away" can become dangerous quickly.



Fog Signals Without Fail

The signals in Rule 35 (every 2 minutes) must never be skipped — it is the only way other vessels can detect you.



The "Radar Plotting Speed" Trap

Continuing at normal speed just because you have radar breaches Rules 6 and 19 — the fog itself limits your safe speed.



PART 5

Lights and Shapes

The rules that let you tell, by day or by night, what kind of vessel you are seeing, where she is heading and what she is doing.



RULE 21 – DEFINITIONS

Types of Light

Every later configuration rule is a combination of these five light types.



Masthead Light

A white light over an arc of 225°, fixed and unbroken, fitted on the fore-and-aft centreline.



Sidelights

Green to starboard, red to port, each over an arc of 112.5°, fixed and unbroken.



Sternlight

A white light over an arc of 135°, fixed and unbroken, placed as near the stern as practicable.



All-Round Light (360°)

A fixed, unbroken light over an arc of 360° — colour varies by rule (white, red or yellow).

RULE 22 – VISIBILITY OF LIGHTS

Minimum Range by Vessel Length

Minimum visibility in nautical miles — longer vessels carry stronger lights.

Vessel Length	Masthead Light	Sidelights	Sternlight	All-Round Light
50m and over	6 nm	3 nm	3 nm	3 nm
12m – 50m	5 nm (3 nm if under 12m)	2 nm	2 nm	2 nm
Under 12m	2 nm	1 nm	2 nm	2 nm

Power-Driven, Towing and Sailing Vessels

The three most common configurations.



Rule 23 – Power-Driven Vessel Underway

Making way: masthead light (a second, higher, aft masthead light if over 50m), sidelights, sternlight.



Rule 24 – Vessel Towing

Carries an additional 2nd masthead light in a vertical line (a 3rd if the length of tow exceeds 200m) and a yellow towing light at the stern.



Rule 25 – Sailing Vessel Underway

Sidelights and sternlight; NO masthead light. May optionally carry two all-round lights at the masthead, red over green.



Small Craft (Under 7m)

The full set if practicable; otherwise at least a white light or lit lantern kept ready to show in time.

Vessels in a Special Activity or Condition

These vessels signal a special condition outside ordinary right-of-way.



Rule 26 – Fishing Vessel

Trawling: green over white all-round light (plus masthead light if making way). Other fishing: red over white all-round light.



Rule 27 – NUC / RAM

NUC: two red all-round lights in a vertical line + two ball shapes by day.
RAM: red-white-red vertical lights + ball-diamond-ball shapes by day.



Rule 28 – Vessel Constrained by Her Draught

Three red all-round lights in a vertical line, or a cylinder shape by day — optional, but an important warning on large vessels.



Rule 29–30 – Pilot Vessel and Anchored/Aground

Pilot: white over red all-round lights. Anchored: one or two (over 50m) white all-round lights + a ball shape by day. Aground: anchor lights + two red vertical lights + three balls by day.

A Seaplane on the Water

Full compliance with this Part may be impracticable for a seaplane; the closest practicable alternative then applies.



General Principle

A seaplane on the water shall, as far as possible, exhibit lights and shapes as closely similar in characteristics and position to those of a power-driven vessel as prescribed in this Part.



When Full Compliance Is Not Possible

If exact compliance is not physically possible, the master applies the alternative arrangement that best conveys the vessel's condition and intentions.



Position in the Hierarchy

In the Rule 18 order of responsibility, the seaplane sits at the very bottom — she gives way to every other vessel type, weather permitting.



Practical Note

For merchant crews this rule is rarely encountered in practice, but it remains examinable background knowledge.



PART 6

Sound and Light Signals

Speaking with the whistle: what equipment each vessel must carry, signalling manoeuvring intent, fog signals and distress signals.



Whistle Signal Durations

Every manoeuvring and fog signal is built from these two basic durations.

Short Blast

A blast of about one second's duration.

Prolonged Blast

A blast of 4 to 6 seconds' duration.

Whistle

Any sound signalling appliance capable of producing the required blasts, as specified technically in Annex III.

RULE 33 – EQUIPMENT FOR SOUND SIGNALS

Mandatory Devices by Vessel Length

The requirement to carry a whistle, bell and gong depends on the vessel's length.

Vessel Length	Whistle	Bell	Gong	Any Sound Device
Under 12m	—	—	—	Mandatory (+ a light source at hand)
12m – 100m	Mandatory	Mandatory	—	—
100m and over	Mandatory	Mandatory	Mandatory (aft)	—

When in Sight of One Another

When you alter course, you tell the other vessel WHAT you are doing.



One Short Blast

"I am altering my course to starboard."



Two Short Blasts

"I am altering my course to port."



Three Short Blasts

"I am operating astern propulsion."



Five or More Short Blasts (Rapid)

"I fail to understand your intentions/actions" or "I doubt whether you are taking sufficient action" — a doubt/danger signal, often accompanied by a light signal.

RULE 35 – SOUND SIGNALS IN RESTRICTED VISIBILITY

At Intervals of No More Than 2 Minutes

In restricted visibility, identity and condition are communicated by sound signal alone.



Power-Driven Vessel Making Way

One prolonged blast.



Power-Driven Vessel Underway But Stopped

Two prolonged blasts in succession, with a short pause between.



NUC / RAM / Constrained by Draught / Sailing / Fishing / Towing

One prolonged followed by two short blasts (a manned vessel being towed adds one further short blast).



At Anchor / Aground

At anchor: rapid ringing of the bell (about 5 seconds) at intervals of not more than 1 minute (a vessel over 100m also sounds a gong aft).
Aground: the same bell signal plus three separate strokes before and after it.

Signals to Attract Attention and Distress Signals

These two final signal rules must be kept distinct from everyday manoeuvring.



Rule 36 – Attracting Attention

Any signal may be used, but it must not be mistaken for an aid to navigation, and high-intensity intermittent (strobe) lights must not be used.



Rule 37 and Annex IV – Distress Signals

MAYDAY (radio), SOS (morse), red star rockets/flares, slowly and repeatedly raising and lowering the arms, orange smoke, the NC flag signal.



Visual Equipment

Red parachute flares, hand-held flares, EPIRB activation, radiotelegraph/telephone alarm signal.



Important Warning

These signals are to be used ONLY to indicate genuine distress/need of assistance — misuse carries serious legal consequences.



PART 7

Exemptions and Summary

A short reminder of Rule 38 and a summary of the whole course.



Transitional Provisions for Existing Vessels

This rule concerns vessels built before the rules entered into force, not newly-built vessels.



Purpose

Gives vessels built before a given date a reasonable transitional period to achieve technical compliance with light/equipment requirements.



Scope

Limited to technical details such as light placement distances and colour characteristics — it does not affect the steering/sailing rules (Part B).



Relevance Today

The exemption periods have long since expired for every vessel in the fleet; this rule is now of historical/reference interest only.



Practical Consequence

When you board a vessel, assume her light configuration is fully COLREG-compliant, and report it if you notice otherwise.

QUICK RECAP

Before You Go on Watch

Ask yourself these three questions at the start of every watch.

Do I see her?

If there is visual contact, Part 3 (Rules 11–18) applies; if not, or if visibility is restricted, Rule 19 applies.

Who Gives Way?

Hierarchy: not under command/RAM > constrained by draught > fishing > sailing > power-driven vessel > seaplane; within the same category, the situation (head-on / crossing / overtaking) decides.

Is My Action Enough?

It must be early, substantial, and clearly noticeable visually or by radar — a small last-minute correction is NOT ENOUGH (Rules 8 & 16).



TRAINING COMPLETE

COLREG is a discipline built on early recognition, correct interpretation, and timely, substantial action — not on luck.

Company DPA — 24/7 emergency line

Fleet Marine Operations Department — mng@gloryship.com.tr

Source: IMO COLREG 1972 (as amended) — the official convention text was used as reference.

